

**33.0.1 NIELIT 2018-50**

The process executes the following code and after execution _____ number of child process get created

```
fork();  
fork();  
fork();  
fork();
```

- A. 4 B. 1 C. 15 D. 16

nielit-2018 operating-system

Answer key

33.0.2 NIELIT 2017 DEC Scientific Assistant A - Section B: 58

Page fault frequency in an operating system is reduced when the :

- A. Process tend to be I/O bound B. Locality of reference is applicable to the process
C. Size of pages is reduced D. Processes tend to be CPU bound

nielit2017dec-assistanta operating-system page-fault

Answer key

33.0.3 NIELIT 2017 DEC Scientific Assistant A - Section B: 48

The process of loading the operating system into memory is called :

- A. Booting B. Spooling C. Thrashing D. Formatting

nielit2017dec-assistanta operating-system

Answer key

33.0.4 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 33

The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page, with 1 free main memory frame is recorded as follows. What is the number of Page Faults?

0100, 0200, 0430, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0370.

- A. 15,4 B. 6,4 C. 7,2 D. 4,6

nielit2017oct-assistanta-cs operating-system demand-paging page-fault

Answer key

33.0.5 NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 10



The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page, with 1 free main memory frame is recorded as follows. What is the number of Page Faults?

0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0370.

- A. 15,4 B. 6,4 C. 7,2 D. 4,6

nielit2017oct-assistanta-it operating-system demand-paging page-fault

Answer key

33.0.6 NIELIT Scientist B 2020 November: 70



On computers where there are multiple operating system, the decision to load a particular one is done by _____.

- A. *PCB* B. Inode C. File Control Block D. Boot Loader

nielit-scb-2020 operating-system

Answer key

33.1 Access Control (1)

33.1.1 Access Control: NIELIT 2021 Dec Scientist B - Section B: 64



The operating system stores an _____ in order to decide to which user to grant which access rights to which file ?

- A. File allocation table B. Process control block
C. Access control matrix D. File control matrix

nielit2021dec-scientistb operating-system access-control matrix

33.2 Bankers Algorithm (2)

33.2.1 Bankers Algorithm: NIELIT 2016 DEC Scientist B (IT) - Section B: 48



The Banker's algorithm is used:

- A. to rectify deadlock. B. to prevent deadlock.
C. to detect deadlock. D. to detect and solve deadlock.

nielit2016dec-scientistb-it operating-system deadlock-prevention-avoidance-detection bankers-algorithm

Answer key

33.2.2 Bankers Algorithm: NIELIT 2022 April Scientist B | Section B | Question: 100



An operating system uses the Banker's algorithm for deadlock avoidance when managing the allocation of three resource types *X*, *Y*, and *Z* to three processes *P*₀, *P*₁, and *P*₂. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type

required by each process during its execution.

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

There are 3 units of type X , 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state:

REQ1: P0 requests 0 units of X , 0 units of Y and 2 units of Z

REQ2: P1 requests 2 units of X , 0 units of Y and 0 units of Z

Which one of the following is TRUE ?

- A. Only REQ1 can be permitted.
- B. Only REQ2 can be permitted.
- C. Both REQ1 and REQ2 can be permitted.
- D. Neither REQ1 nor REQ2 can be permitted.

nielit2022apr-scientistb operating-system deadlock-prevention-avoidance-detection bankers-algorithm

33.3

CO and Architecture (2)

33.3.1 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 110



In _____ machine is executing operating system instruction :

- A. System mode
- B. User mode
- C. Normal mode
- D. Safe mode

nielit2021dec-scientistb operating-system co-and-architecture

Answer key

33.3.2 CO and Architecture: NIELIT 2021 Dec Scientist B - Section B: 45



Which of the following is true?

- A. Unless enabled, a CPU will not be able to process interrupts
- B. Loop instructions cannot be interrupted till they complete
- C. A processor need not checks for interrupts before executing a new instruction
- D. Only level triggered interrupts are possible on microprocessors

nielit-2021-it-dec-scientistb co-and-architecture operating-system

Answer key

33.4

Cache Memory (2)

33.4.1 Cache Memory: NIELIT 2016 DEC Scientist B (CS) - Section B: 37



The principle of locality of reference justifies the use of

- A. Non reusable
- B. Cache memory
- C. Virtual memory
- D. None of the above

nielit2016dec-scientistb-cs operating-system cache-memory

Answer key

33.4.2 Cache Memory: NIELIT 2017 DEC Scientific Assistant A - Section B: 56



In a particular system it is observed that, the cache performance gets improved as a result of increasing the block size of the cache. The primary reason behind this is :

- A. Programs exhibits temporal locality
- B. Programs have small working set
- C. Read operation is frequently required rather than write operation
- D. Programs exhibits spatial locality

nielit2017dec-assistanta operating-system cache-memory

Answer key

33.5

Context Switch (1)

33.5.1 Context Switch: NIELIT 2017 DEC Scientist B - Section B: 17



The time taken to switch between user and kernel modes of execution be t_1 while the time taken to switch between two processes be t_2 . Which of the following is TRUE?

- A. $t_1 > t_2$
- B. $t_1 = t_2$
- C. $t_1 < t_2$
- D. nothing can be said about the relation between t_1 and t_2

nielit2017dec-scientistb operating-system context-switch

Answer key

33.6

Disk (7)

33.6.1 Disk: NIELIT 2016 MAR Scientist B - Section C: 36



What is the elapsed time of P if records of F are organized using a blocking factor of 2 (i.e. each block on D contains two records of F) and P uses one buffer?

- A. 12 sec.
- B. 14 sec.
- C. 17 sec.
- D. 21 sec.

nielit2016mar-scientistb operating-system disk

33.6.2 Disk: NIELIT 2016 MAR Scientist B - Section C: 8



When we move from the outermost track to the innermost track in a magnetic disk, then density(bits per linear inch)

- A. increases.
- B. decreases.
- C. remains the same.
- D. either remains constant or decreases.

nielit2016mar-scientistb operating-system disk

Answer key

33.6.3 Disk: NIELIT 2016 MAR Scientist C - Section C: 26



The seek time of a disk is 30 ms. It rotates at the rate of 30 rotations/second. The capacity of each track is 300 words. The access time is (approximately)

- A. 62 ms
- B. 60 ms
- C. 50 ms
- D. 47 ms

nielit2016mar-scientistc operating-system disk

Answer key

33.6.4 Disk: NIELIT 2017 July Scientist B (CS) - Section B: 25



Which access method is used for obtaining a record from cassette tape?

- A. Direct
- B. Sequential
- C. Random
- D. Parallel

nielit2017july-scientistb-cs operating-system disk

Answer key

33.6.5 Disk: NIELIT 2018-52



In the disk, swap space is used to _____.

- A. save XML files
- B. save process data
- C. save drivers
- D. save HTML files

nielit-2018 operating-system disk

Answer key

33.6.6 Disk: NIELIT 2022 April Scientist B | Section B | Question: 118



An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10 ms. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected)

- A. 0.50 s
- B. 1.50 s
- C. 1.25 s
- D. 1.00 s

nielit2022apr-scientistb operating-system disk co-and-architecture

33.6.7 Disk: NIELIT 2022 April Scientist B | Section B | Question: 45

Consider a hard disk with 16 recording surfaces (0 – 15) having 16384 cylinders (0 – 16383) and each cylinder contains 64 sectors (0 – 63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder number, surface number, sector number>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is < 1200, 9, 40 >. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

- A. 1281 B. 1282 C. 1283 D. 1284

nielit2022apr-scientistb operating-system disk normal

33.7**Disk Scheduling (4)****33.7.1 Disk Scheduling: NIELIT 2017 July Scientist B (CS) - Section B: 35**

A disk has 200 tracks (numbered 0 through 199). At a given time, it was servicing the request of reading data from track 120, and at the previous request, service was for track 90. The pending requests (in order of their arrival) are for track numbers.

30 70 115 130 110 80 20 25

How many times will the head change its direction for the disk scheduling policies SSTF (Shortest Seek Time First) and FCFS (First Come First Serve)?

- A. 2 and 3 B. 3 and 3 C. 3 and 4 D. 4 and 4

nielit2017july-scientistb-cs operating-system disk-scheduling

Answer key

33.7.2 Disk Scheduling: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 35

Disk request come to a disk driver for cylinders in the order 10, 22, 20, 2, 40, 6 and 38, at a time when the disk drive is reading from cylinder 20. The seek time is 6 ms per cylinder. The total seek time, if the disk arm scheduling algorithm is first-come-first-served is

- A. 900 ms B. 850 ms C. 360 ms D. 876 ms

nielit2017oct-assistanta-cs operating-system disk-scheduling

Answer key

33.7.3 Disk Scheduling: NIELIT 2021 Dec Scientist B - Section B: 70

Consider a memory system where a request comes for disk driver for cylinders 10, 22, 20, 2, 40, 6 and 38 (head start at 20). The seek time is 6 ms per cylinder. The total seek time if the disk arm scheduling algorithm is FCFS :

- A. 850 ms B. 906 ms C. 400 ms D. 876 ms

33.7.4 Disk Scheduling: NIELIT Scientist B 2020 November: 72

One disk queue with requests for I/O to blocks on cylinders. The Request are in the following manner:

98 183 37 122 14 124 65 67

Considering $SSTF$ (shortest seek time first) scheduling, the total number of head movements is, if the disk head is initially at 53 is:

- A. 236 B. 246 C. 220 D. 240

nielit-scb-2020 process-scheduling operating-system disk-scheduling

Answer key

33.8 Effective Memory Access (1)**33.8.1 Effective Memory Access: NIELIT Scientist B 2020 November: 85**

If main memory access time is $400 \mu s$, TLB access time $50 \mu s$, considering TLB hit 90%, what will be the overall access time?

- A. $800 \mu s$ B. $490 \mu s$ C. $485 \mu s$ D. $450 \mu s$

nielit-scb-2020 virtual-memory effective-memory-access operating-system

Answer key

33.9 File Organization (2)**33.9.1 File Organization: NIELIT 2016 MAR Scientist B - Section C: 35**

The file structure that redefines its first record at a base of zero uses the term

- A. relative organization. B. key fielding.
C. dynamic reallocation. D. all of these.

nielit2016mar-scientistb operating-system file-system file-organization

Answer key

33.9.2 File Organization: NIELIT 2017 DEC Scientific Assistant A - Section B: 47

The open file table has a/an _____ associated with each file.

- A. file content B. file permission C. open count D. close count

nielit2017dec-assistanta operating-system file-organization

Answer key

33.10 File System (3)

33.10.1 File System: NIELIT 2016 MAR Scientist C - Section C: 2



The file manager is responsible for

- A. naming files
- B. saving files
- C. deleting files
- D. all of these

nielit2016mar-scientistc operating-system file-system

Answer key 

33.10.2 File System: NIELIT 2017 DEC Scientific Assistant A - Section B: 7



If file size is large and if it is to be accessed randomly then which of the following allocation strategy should be best to use in a system?

- A. Linked allocation
- B. Indexed allocation
- C. Contiguous allocation
- D. None of the options

nielit2017dec-assistanta operating-system file-system

Answer key 

33.10.3 File System: NIELIT Scientist B 2020 November: 104



Which table is used in MS DOS for linked list allocation?

- A. *TLB*
- B. Page Table
- C. *FAT*
- D. Index Table

nielit-scb-2020 file-system linked-list

Answer key 

33.11 Fragmentation (2)

33.11.1 Fragmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 29



What is Compaction ?

- A. a technique for overcoming internal fragmentation
- B. a paging technique
- C. a technique for overcoming external fragmentation
- D. a technique for overcoming fatal error

nielit2017dec-assistanta operating-system fragmentation

Answer key 

33.11.2 Fragmentation: NIELIT Scientist B 2020 November: 49



Contiguous memory allocation having variable size partition suffers from:

- A. External Fragmentation
- B. Internal Fragmentation
- C. Both External and Internal Fragmentation
- D. None of the options

nielit-scb-2020 operating-system fragmentation memory-management

Answer key 

33.12

Memory Management (9)

33.12.1 Memory Management: NIELIT 2016 DEC Scientist B (IT) - Section B: 51

Copying a process from memory to disk to allow space for other processes is called:



- A. Swapping B. Demand Paging C. Deadlock D. Page Fault

nielit2016dec-scientistb-it operating-system memory-management

Answer key 

33.12.2 Memory Management: NIELIT 2016 MAR Scientist C - Section C: 8

If there are 32 segments, each of size 1 K byte, then the logical address should have



- A. 13 bits B. 14 bits
C. 15 bits D. 16 bits

nielit2016mar-scientistc operating-system memory-management

Answer key 

33.12.3 Memory Management: NIELIT 2017 DEC Scientist B - Section B: 16

Which of the following is added to the page table in order to track whether a page of cache has been modified since it was read from the memory?



- A. Reference bit B. Dirty bit C. Tag bit D. Valid bit

nielit2017dec-scientistb operating-system memory-management paging cache-memory

Answer key 

33.12.4 Memory Management: NIELIT 2017 July Scientist B (CS) - Section B: 32

A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a Translation Look-aside Buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is



- A. 11 bits B. 13 bits C. 15 bits D. 20 bits

nielit2017july-scientistb-cs operating-system memory-management paging translation-lookaside-buffer

Answer key 

33.12.5 Memory Management: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 4

If a processor has 32-bit virtual address, 28-bit physical address, 2 kb pages. How many bits are required for the virtual, physical page number?



- A. 17,21 B. 21,17 C. 6,10 D. None

nielit2017oct-assistanta-cs operating-system memory-management

Answer key 

33.12.6 Memory Management: NIELIT 2018-38



Find the effective access time for the memory for the given data.

Page fault service time = 8 ms

Average memory access time = 20 ns

One page fault is generated for every memory access = 10^6

- A. 29 ns B. 33 ns C. 28 ns D. 30 ns

nielit-2018 operating-system memory-management

Answer key 

33.12.7 Memory Management: NIELIT 2021 Dec Scientist B - Section B: 69



If the size of the logical address space is 2^m and the page size is 2^n addressing units, then the high order m-n bits of a logical address designate the _____.

- A. offset B. page no C. frame no D. physical address

nielit2021dec-scientistb operating-system memory-management

Answer key 

33.12.8 Memory Management: NIELIT 2021 Dec Scientist B - Section B: 76



In a paging scheme if page size is of 2048 bytes, then while accommodating a process of 72,766 bytes, how much internal fragmentation occurs ?

- A. 962 bytes B. 2048 bytes
C. 1024 bytes D. 1086 bytes

nielit2021dec-scientistb operating-system memory-management

Answer key 

33.12.9 Memory Management: NIELIT Scientific Assistant A 2020 November: 70



A computer has a single cache (off-chip) with a 3 ns hit time and a 95% hit rate. Main memory has a 50 ns access time. If we add an on-chip cache with a 0.6 ns hit time and a 98% hit rate, the computer's effective access time:

- A. 2.8 ns B. 5.5 ns
C. 0.7 ns D. None of the options

nielit-sta-2020 operating-system memory-management

Answer key 

33.13.1 Monitors: NIELIT 2016 MAR Scientist B - Section C: 37



A long-term monitor

- A. should show any immediate performance problems
- B. should show I/O, paging and processor activity
- C. need show only the I/O and processor activity
- D. usually reports only on terminal displays

nielit2016mar-scientistb operating-system monitors

Answer key

33.14.1 Page Replacement: NIELIT 2016 DEC Scientist B (CS) - Section B: 48



In which one of the following pages replacement policies, Belady's anomaly may occur?

- A. FIFO
- B. LRU
- C. Optimal
- D. MRU

nielit2016dec-scientistb-cs operating-system page-replacement

Answer key

33.14.2 Page Replacement: NIELIT 2016 MAR Scientist B - Section C: 38



Determine the number of page faults when references to pages occur in the following order: 1, 2, 4, 5, 2, 1, 2, 4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 having been brought earlier than page 2. (LRU algorithm is used).

- A. 3
- B. 5
- C. 4
- D. None of these.

nielit2016mar-scientistb operating-system page-replacement page-fault least-recently-used

Answer key

33.14.3 Page Replacement: NIELIT 2016 MAR Scientist B - Section C: 39



Working set(t, k) at an instant of time, t , is

- A. The set of k future references that the operating system will make
- B. The set of future references that the operating system will make in the next ' k ' time units
- C. The set of k references with high frequency
- D. The set of pages that have been referenced in the last k time units

nielit2016mar-scientistb operating-system page-replacement

Answer key

33.14.4 Page Replacement: NIELIT 2017 DEC Scientist B - Section B: 51



The total number of page faults for the reference string 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 using FIFO page replacement policy for a process, if 3 frames are allocated to it are

- A. 9 B. 10 C. 8 D. 11

nielit2017dec-scientistb operating-system page-replacement page-fault

Answer key

33.14.5 Page Replacement: NIELIT 2021 Dec Scientist A - Section B: 102



Consider a system with page size p and average process size m and size of each page table entry is e . What is the amount of space required by page table ?

- A. me/p B. mp/e C. mpe D. pe/m

nielit2021dec-scientista operating-system page-replacement memory-management

Answer key

33.14.6 Page Replacement: NIELIT 2021 Dec Scientist A - Section B: 107



Consider a system with three frames in memory and following memory references in the working set

2 1 2 3 5 4 1 3 4 2 1

How many page fault will be there if we use second chance page replacement algorithm?

- A. 7 B. 8 C. 9 D. 10

nielit2021dec-scientista operating-system page-replacement memory-management

33.14.7 Page Replacement: NIELIT 2021 Dec Scientist A - Section B: 114



Which of the following Page Replacement Algorithm suffers from the Belady's anomaly ?

- A. LRU B. Optimal page Replacement
C. FIFO D. Both LRU and FIFO

nielit2021dec-scientista operating-system page-replacement

Answer key

33.14.8 Page Replacement: NIELIT 2022 April Scientist B | Section B | Question: 78



Consider the virtual page reference string

1, 2, 3, 2, 4, 1, 3, 2, 4, 1

On a demand paged virtual memory system running on a computer system that main memory size of 3 pages frames which are initially empty. Let LRU, FIFO and

A. OPTIMAL < LRU < FIFO
C. OPTIMAL = LRU

B. OPTIMAL < FIFO < LRU
D. OPTIMAL = FIFO



Which of the following Page Replacement Algorithm suffers from the belady's anomaly?

A. *LRU*
C. *FIFO*

B. Optimal Page Replacement
D. Both *LRU* and *FIFO*

Answer key 

Process (2)

A. 0% B. 10.6% C. 30.0% D. 89.4%



A. Zombies B. Orphans C. Parent Process D. Child Process

Answer key

Process Scheduling (13)

The degree of multi programming is controlled by:

- A. CPU Scheduler
- C. Context Switching

- B. Long-term Scheduler
- D. Medium term Scheduler

nielit2016dec-scientistb-it operating-system process-scheduling

Answer key 

33.16.2 Process Scheduling: NIELIT 2016 MAR Scientist C - Section C: 7



In real-time operating systems, which of the following is the most suitable scheduling scheme?

- A. round-robin
- C. preemptive
- B. first-come-first-served
- D. random scheduling

nielit2016mar-scientistc operating-system process-scheduling

Answer key 

33.16.3 Process Scheduling: NIELIT 2017 DEC Scientific Assistant A - Section B: 34



Process is in a ready state _____ .

- A. when process is scheduled to run after some execution
- B. when process is unable to run until some task has been completed
- C. when process is using the *CPU*
- D. none of the options

nielit2017dec-assistanta operating-system process-scheduling

Answer key 

33.16.4 Process Scheduling: NIELIT 2017 DEC Scientific Assistant A - Section B: 57



Starvation can be avoided by which of the following statements:

- i. By using shortest job first resource allocation policy .
- ii. By using first come first serve resources allocation policy.

- A. (i) only
- B. (i) and (ii) only
- C. (ii) only
- D. None of the options

nielit2017dec-assistanta operating-system process-scheduling

Answer key 

33.16.5 Process Scheduling: NIELIT 2017 DEC Scientist B - Section B: 12



Consider the following four processes with their corresponding arrival time and burst time:

Process	Arrival time	Burst time(in ms)
P1	0.0	8
P2	0.6	6
P3	3.8	4
P4	4.4	2

What is the average turn around time (in ms) for these processes using FCFS scheduling algorithm?

- A. 15 B. 12.8 C. 13 D. none of the options

nielit2017dec-scientistb operating-system process-scheduling

Answer key 

33.16.6 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 44



Consider three processes (process id 0, 1, 2 respectively) with compute time bursts 2, 4 and 8 time units. All processes arrive at time zero. Consider the Longest Remaining Time First (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is

- A. 13 units B. 14 units C. 15 units D. 16 units

nielit2017july-scientistb-it operating-system process-scheduling

Answer key 

33.16.7 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 45



Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

- A. 0% B. 10.6% C. 30.0% D. 89.4%

nielit2017july-scientistb-it operating-system process-scheduling

Answer key 

33.16.8 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 46



Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

A. 1

B. 2

C. 3

D. 4

nielit2017july-scientistb-it operating-system process-scheduling

Answer key 

33.16.9 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 47



Which of the following process scheduling algorithm may lead to starvation?

A. FIFO

B. Round Robin

C. Shortest Job Next

D. None of the option

nielit2017july-scientistb-it operating-system process-scheduling

Answer key 

33.16.10 Process Scheduling: NIELIT 2017 July Scientist B (IT) - Section B: 48



A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

A. This algorithm is equivalent to the first come first serve algorithm.

B. This algorithm is equivalent to the round-robin algorithm.

C. This algorithm is equivalent to the shortest-job-first algorithm.

D. This algorithm is equivalent to the shortest-remaining time-first algorithm.

nielit2017july-scientistb-it operating-system process-scheduling

Answer key 

33.16.11 Process Scheduling: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 9



Which of the following scheduling algorithms could result in starvation?

A. Priority

B. Round Robin

C. FCFS

D. None of the above

nielit2017oct-assistanta-it operating-system process-scheduling

Answer key 

33.16.12 Process Scheduling: NIELIT 2018-78



Identify the true statement from the given statements.

1. FIFO is non-preemptive

2. Round robin is non-preemptive

3. Multilevel Queue Scheduling is non-preemptive

A. 1

B. 1 and 2

C. 1, 2 and 3

D. 2

nielit-2018 operating-system process-scheduling

Answer key 

33.16.13 Process Scheduling: NIELIT 2021 Dec Scientist B - Section B: 57



_____ is the elapsed time between the time a program or job is submitted and the time when it is completed.

- A. Response time
- B. Turnaround time
- C. Waiting time
- D. Throughput

nielit2021dec-scientistb operating-system process-scheduling

Answer key 

33.17

Process Synchronization (2)

33.17.1 Process Synchronization: NIELIT 2016 DEC Scientist B (CS) - Section B: 51

A process that is based on IPC mechanism which executes on different systems and can communicate with other processes using message based communication is called _____.



- A. Local Procedure Call
- B. Remote Procedure Call
- C. Inter Process Communication
- D. Remote Machine Invocation

nielit2016dec-scientistb-cs operating-system process-synchronization

Answer key 

33.17.2 Process Synchronization: NIELIT 2016 DEC Scientist B (IT) - Section B: 50

Which of the following statements about semaphores is true?



- A. P and V operations should be indivisible operations.
- B. A semaphore implementation should guarantee that threads do not suffer indefinite postponement.
- C. If several threads attempt a $P(S)$ operation simultaneously, only one thread should be allowed to proceed.
- D. All of the above.

nielit2016dec-scientistb-it operating-system process-synchronization

Answer key 

33.18

Process and Threads (2)

33.18.1 Process and Threads: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 23



Which two are valid constructions for Thread?

1. Thread(Runnable r, String name)
2. Thread()
3. Thread(int priority)

4. Thread(Runnable r, ThreadGroup g)
5. Thread(Runnable r, int priority)

A. 1 and 3 B. 2 and 4 C. 1 and 2 D. 2 and 5

nielit2017oct-assistanta-it operating-system process-and-threads threads

Answer key 

33.18.2 Process and Threads: NIELIT 2021 Dec Scientist A - Section B: 120



Which one of the following cannot be scheduled by the kernel ?

- | | |
|------------------------|-----------------------|
| A. Kernel level thread | B. User level thread |
| C. Process | D. None of the option |

nielit2021dec-scientista operating-system process-scheduling process-and-threads

Answer key 

33.19 Resource Allocation (6)

33.19.1 Resource Allocation: NIELIT 2016 DEC Scientist B (CS) - Section B: 56



Consider a system with m resources of same type being shared by n processes. Resources can be requested and released by processes only one at a time. The system is deadlock free if and only if

- | | |
|---|---|
| A. The sum of all max needs is
$< m + n$ | B. The sum of all max needs is
$> m + n$ |
| C. Both of above | D. None |

nielit2016dec-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key 

33.19.2 Resource Allocation: NIELIT 2017 DEC Scientist B - Section B: 21



A system has 3 processes sharing 4 resources. If each process needs a maximum of 2 units then, deadlock

- | | |
|--------------------|------------------------|
| A. Can never occur | B. Has to occur |
| C. May occur | D. None of the options |

nielit2017dec-scientistb operating-system resource-allocation

Answer key 

33.19.3 Resource Allocation: NIELIT 2017 July Scientist B (CS) - Section B: 36



Consider the following snapshot of a system running n processes. Process i is holding X_i instances of a resource R , $1 \leq i \leq n$. Currently, all instances of R are occupied. Further, for all i , process i has placed a request for an additional Y_i instances while holding the X_i instances it already has. There are exactly two processes p and q and such that $Y_p = Y_q = 0$. Which one of the following can serve as a necessary condition to guarantee that the system is not approaching a deadlock?

- A. $\min(X_p, X_q) < \max(Y_i)$ where $i! = p$ and $i! = q$ and $X_p - q X_q \geq \min(Y_i)$ where $i! = p$ and $i! = q$
 C. $\max(X_p, X_q) > 1$ D. $\min(X_p, X_q) > 1$

nielit2017july-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.19.4 Resource Allocation: NIELIT 2017 July Scientist B (CS) - Section B: 37

A system has n resources $R_0, \dots, R_{n-1}$, and k processes $P_0, \dots, P_{k-1}$. The implementation of the resource request logic of each process P_i is as follows:



```
if(i%2==0){
    if(i<n) request Ri;
    if(i+2<n) request Ri+2;
}
else{
    if(i<n) request Rn-i;
    if(i+2<n) request Rn-i-2;
}
```

In which of the following situations is a deadlock possible?

- A. $n = 40, k = 26$ B. $n = 21, k = 12$ C. $n = 20, k = 10$ D. $n = 41, k = 19$

nielit2017july-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.19.5 Resource Allocation: NIELIT 2017 July Scientist B (CS) - Section B: 38

A system contains three programs and each requires three tape units for its operation. The minimum number of tape units which the system must have such that deadlocks never arise is _____.



- A. 6 B. 7 C. 8 D. 9

nielit2017july-scientistb-cs operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.19.6 Resource Allocation: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 15

Consider a system having ' m ' resources of the same type. These resources are shared by 3 processes A, B, C ; which have peak time demands of 3, 4, 6 respectively. The minimum value of ' m ' that ensures that deadlock will never occur is



- A. 11 B. 12 C. 13 D. 14

nielit2017oct-assistanta-it operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key

33.20.1 Segmentation: NIELIT 2017 DEC Scientific Assistant A - Section B: 17



Non-contiguous memory allocation splits program into blocks of memory called _____ that can be loaded in non-adjacent holes in main memory.

- A. Pages B. Frames C. Partition D. Segments

nielit2017dec-assistanta operating-system page-replacement segmentation

Answer key

33.21

Semaphore (3)

33.21.1 Semaphore: NIELIT 2018-57



In a system, counting semaphore was initialized to 10, then $6P(\text{wait})$ operations and $4V(\text{signal})$ operations were completed on this semaphore. So _____ is the final value of the semaphore.

- A. 7 B. 8 C. 13 D. 12

nielit-2018 operating-system semaphore

Answer key

33.21.2 Semaphore: NIELIT 2021 Dec Scientist B - Section B: 79



A protected variable which can be accessed and changed by particular set of operation is called :

- A. interrupt B. monitor C. semaphore D. IPC

nielit2021dec-scientistb operating-system process-synchronization semaphore

Answer key

33.21.3 Semaphore: NIELIT 2022 April Scientist B | Section B | Question: 110



The atomic *fetch-and-set* x, y instruction unconditionally sets the memory location x to 1 and fetches the old value of x in y without allowing any intervening access to the memory location x . Consider the following implementation of P and V functions on a binary semaphore S .

```
void P (binary_semaphore *s) {
    unsigned y;
    unsigned *x = &(s->value);
    do {
        fetch-and-set x, y;
    } while (y);
}
```

```
void V (binary_semaphore *s) {
    S->value = 0;
}
```

Which one of the following is true?

- A. The implementation may not work if context switching is disabled in P
- B. Instead of using *fetch-and-set*, a pair of normal load/store can be used
- C. The implementation of V is wrong
- D. The code does not implement a binary semaphore

nielit2022apr-scientistb operating-system process-synchronization semaphore

33.22

System Calls (1)

33.22.1 System Calls: NIELIT 2017 DEC Scientific Assistant A - Section B: 37



How many times the word "PROCESS" will be printed when executing the following program ?

```
main(){  
    printf("PROCESS");  
    fflush();  
    fork();  
    fork();  
}
```

- A. 8 B. 4 C. 6 D. 7

nielit2017dec-assistanta operating-system system-calls

Answer key

33.23

Threads (2)

33.23.1 Threads: NIELIT 2017 DEC Scientific Assistant A - Section B: 9



Operating System maintains the page table for :

- A. each process B. each thread
C. each instruction D. each address

nielit2017dec-assistanta operating-system paging threads

Answer key

33.23.2 Threads: NIELIT 2021 Dec Scientist B - Section B: 40



Choose the correct sequence of items mentioned in **List-1** out of items mentioned in **List-2** :

List-1	List-2
(a) Threads	(i) operating system
(b) Non pre-emptive	(ii) first come first serve
(c) Pre-emptive	(iii) round robin algorithm

(d) Scheduling	(iv) Context switch
(e) Process switch	(v) light weight process with in reduced time

- | | | | | | |
|-----|------|-------|-------|------|------|
| | (a) | (b) | (c) | (d) | (e) |
| (A) | (v) | (iii) | (iv) | (ii) | (i) |
| (B) | (ii) | (iii) | (v) | (iv) | (i) |
| (C) | (v) | (ii) | (iii) | (i) | (iv) |
| (D) | (ii) | (iv) | (v) | (i) | (ii) |

nielit-2021-it-dec-scientistb nielit2021dec-scientistb operating-system process-scheduling threads process-synchronization

Answer key 

33.24

Virtual Memory (6)

33.24.1 Virtual Memory: NIELIT 2017 July Scientist B (CS) - Section B: 33



A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table ($T1$), which occupies exactly one page. Each entry of $T1$ stores the base address of a page of the second-level table ($T2$). Each entry of $T2$ stores the base address of a page of the third-level table ($T3$). Each entry of $T3$ stores a Page Table Entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16 way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

What is the size of a page in KB in this computer?

- A. 2 B. 4 C. 8 D. 16

nielit2017july-scientistb-cs operating-system virtual-memory

Answer key 

33.24.2 Virtual Memory: NIELIT 2017 July Scientist B (CS) - Section B: 34



A computer uses 46 – bit virtual address, 32 – bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table ($T1$), which occupies exactly one page. Each entry of $T1$ stores the base address of a page of the second-level table ($T2$). Each entry of $T2$ stores the base address of a page of the third-level table ($T3$). Each entry of $T3$ stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16 way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

What is the minimum number of page colours needed to guarantee that no two synonyms

map to different sets in the processor cache of this computer?

- A. 2 B. 4 C. 8 D. 16

nielit2017july-scientistb-cs operating-system virtual-memory

33.24.3 Virtual Memory: NIELIT 2018-48



Decreasing the RAM causes

- A. fewer page faults B. more page faults
C. virtual memory gets increased D. virtual memory gets decreased

nielit-2018 operating-system virtual-memory

Answer key

33.24.4 Virtual Memory: NIELIT 2021 Dec Scientist B - Section B: 42



What is virtual memory?

- A. An illusion of an extremely large memory, that is physically not present
B. A type of memory used in nano computers
C. A memory that is in direct touch of CPU
D. A memory that works on the principle of set associative mapping

nielit-2021-it-dec-scientistb operating-system virtual-memory

Answer key

33.24.5 Virtual Memory: NIELIT 2021 Dec Scientist B - Section B: 53



A page fault occurs when:

- A. the Deadlock happens B. the Segmentation starts
C. the page is found in the memory D. the page is not found in the memory

nielit-2021-it-dec-scientistb operating-system virtual-memory

Answer key

33.24.6 Virtual Memory: NIELIT 2021 Dec Scientist B - Section B: 77



Consider a memory system in which the size of page is 8 KB. A CPU that generates the 32 bit virtual address. What will be minimum size of TLB (translation look aside buffer) tag, if TLB has total 256 page table and is 8 way set associative?

- A. 12 bit B. 13 bit C. 14 bit D. 15 bit

nielit-2021-it-dec-scientistb co-and-architecture memory-management virtual-memory

Answer key

Answer Keys

33.0.1	C	33.0.2	B	33.0.3	A	33.0.4	C	33.0.5	C
33.0.6	D	33.1.1	C	33.2.1	B	33.2.2	B	33.3.1	A
33.3.2	A	33.4.1	B	33.4.2	D	33.5.1	C	33.6.1	TBA
33.6.2	A	33.6.3	D	33.6.4	B	33.6.5	B	33.6.6	B
33.6.7	D	33.7.1	C	33.7.2	D	33.7.3	D	33.7.4	A
33.8.1	B	33.9.1	A	33.9.2	C	33.10.1	D	33.10.2	B
33.10.3	C	33.11.1	C	33.11.2	A	33.12.1	A	33.12.2	C
33.12.3	B	33.12.4	C	33.12.5	B	33.12.6	C	33.12.7	B
33.12.8	A	33.12.9	C	33.13.1	B	33.14.1	A	33.14.2	C
33.14.3	D	33.14.4	B	33.14.5	A	33.14.6	C	33.14.7	C
33.14.8	B	33.14.9	C	33.15.1	B	33.15.2	A	33.16.1	B
33.16.2	C	33.16.3	A	33.16.4	C	33.16.5	B	33.16.6	A
33.16.7	B	33.16.8	B	33.16.9	C	33.16.10	B	33.16.11	A
33.16.12	A	33.16.13	B	33.17.1	B	33.17.2	D	33.18.1	C
33.18.2	B	33.19.1	A	33.19.2	A	33.19.3	B	33.19.4	B
33.19.5	A	33.19.6	A	33.20.1	D	33.21.1	B	33.21.2	C
33.21.3	A	33.22.1	X	33.23.1	A	33.23.2	C	33.24.1	C
33.24.2	C	33.24.3	B	33.24.4	A	33.24.5	D	33.24.6	D